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# Far Field Radiation Transport from Little Boy and Fat Man

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# Far Field Radiation Transport from Little Boy and Fat Man

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## INTRODUCTION

Atomic explosions produce prompt and delayed emissions of neutrons and photons into the atmosphere. The particles move through space and undergo interactions that make up radiation transport. MCNP, a radiation transport code, can simulate this physics with high fidelity [1].

**Purpose:** This body of work aims to recreate neutron dose calculations as a function of radius at ground level for the wartime bombings of Japan (Fat Man and Little Boy) using modern 3D MCNP simulations. The computed fluence results will be compared against previous multi-laboratory simulation efforts to validate contemporary radiation transport modeling for survivability calculations.

## METHODS

### Source Term Modeling

Neutrons are emitted isotopically front a point source located at the burst height. The source term is derived from the Standard Unclassified Outputs paper[2], with units converted to create a time-independent source compatible with MCNP input requirements. The source defines the quantities and intensities of particles produced during the atomic explosion while neglecting angular dependence.

# Neutron Output Range [#/(kt-s)] → [mol/MeV/kt] → Graph and compare to Dosimetry System 2002 [3]

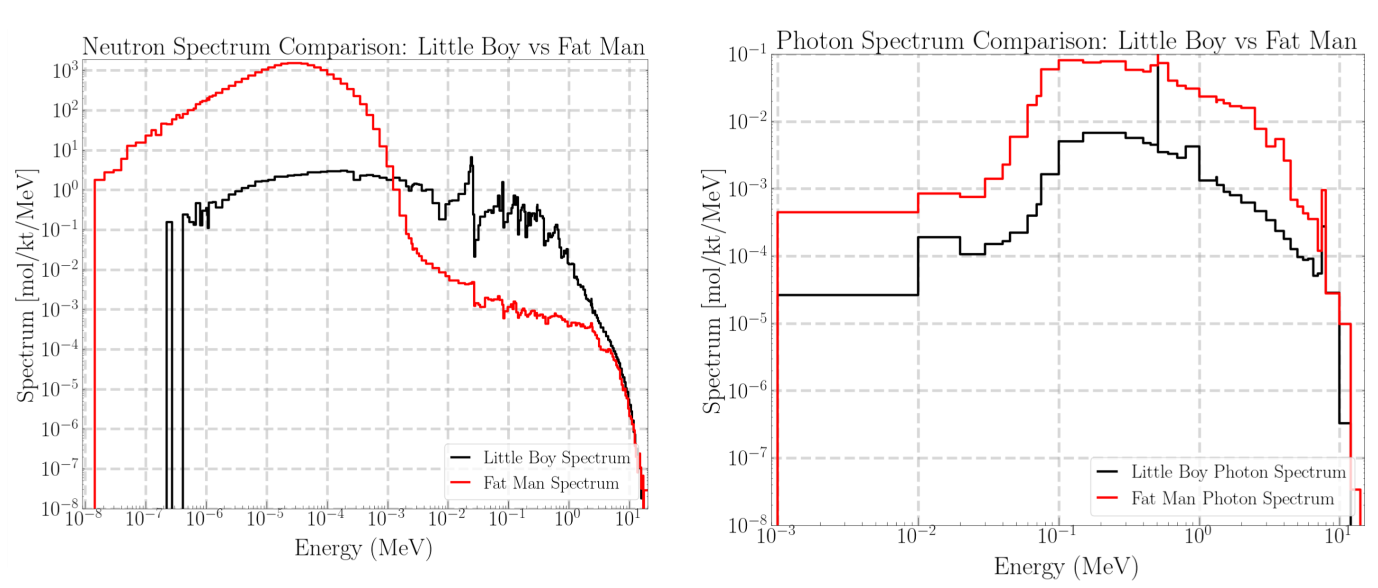
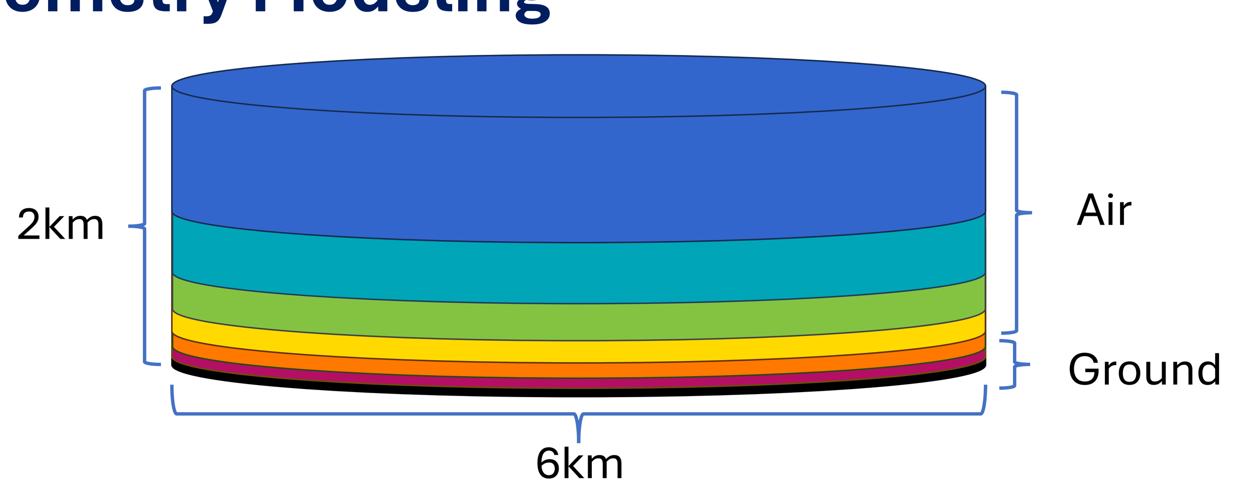



*Figure 1. Neutron and photon energy binned spectrum for Fat Mat and Little Boy*

### Geometry Modeling




Material compositions are taken from Dosimetry System 2002 which represent samples of the ground and air from Nagasaki and Hiroshima [3]

## Variance Reduction

Utilized three variance reduction techniques to reduce uncertainty of the fluence, which is derived from tallies, due to the large domain.

| Source Biasing | Mesh Based Weight Windows | Semi-deterministic Transport |
|---|---|---|
| sample all source energies equally | divide the geometry into small sections and generate more particles in desired areas | assume contribution from interactions that could redirect to a desired point |

**Mesh Based Weight Windows**

Illustration of implementing mesh based weight windows to improve number of neutrons transported to fluence tallies that are kilometers away. This was done by seeding previous weight window generation simulations to reduce statistical error at larger radii.

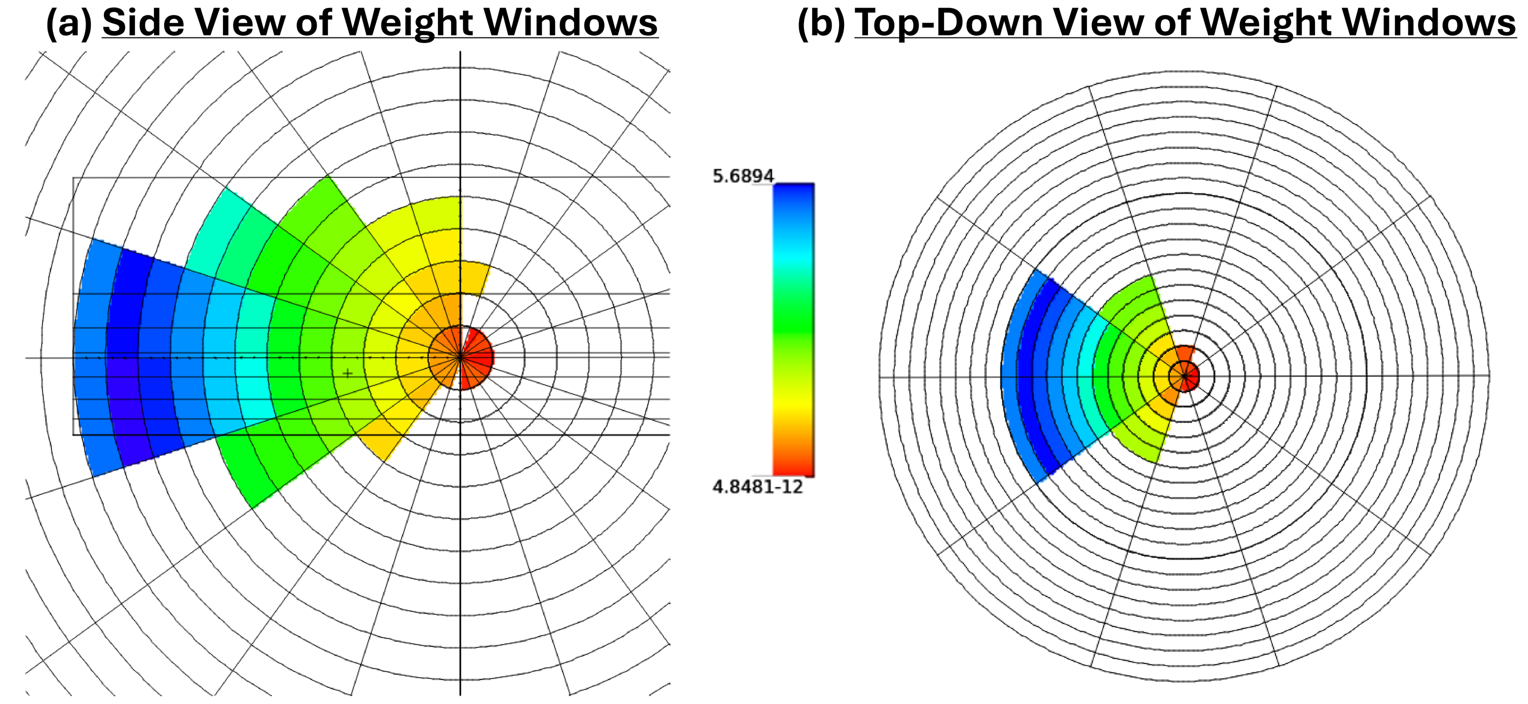



*Figure 2: (a) The tallies are represented by the small tick marks every 100 meters. The weight windows are superimposed over the geometry of the problem. Particles are assigned an importance depending on the weight of the particles. (b) similar representation of particle importance at burst height*

## RESULTS

Fluence is calculated from MCNP simulations to verify against the Dosimetry System 2002 results to build confidence in the model set up. Using above methods, we compare MCNP model to the DS02 paper in two ways: (1) the fluence profile vs altitude and radius (2) fluence vs radius at height of burst

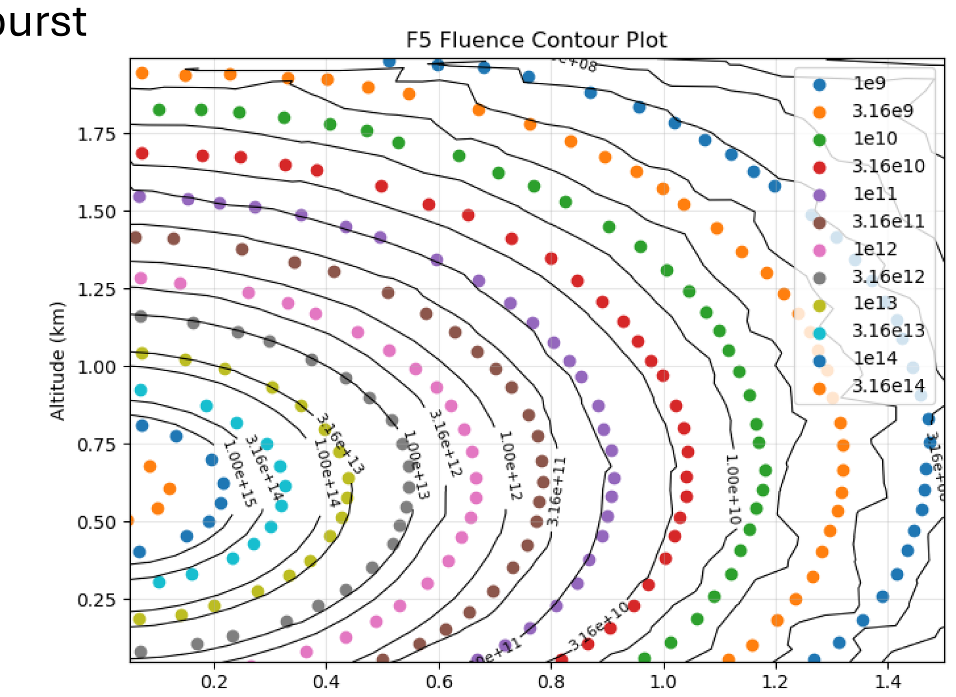



*Figure 2. Neutron fluence contour plots from Little Boy. Calculated with semi-deterministic F5 Tallies in MCNP and compared against Dosimetry System 2002 [2]*

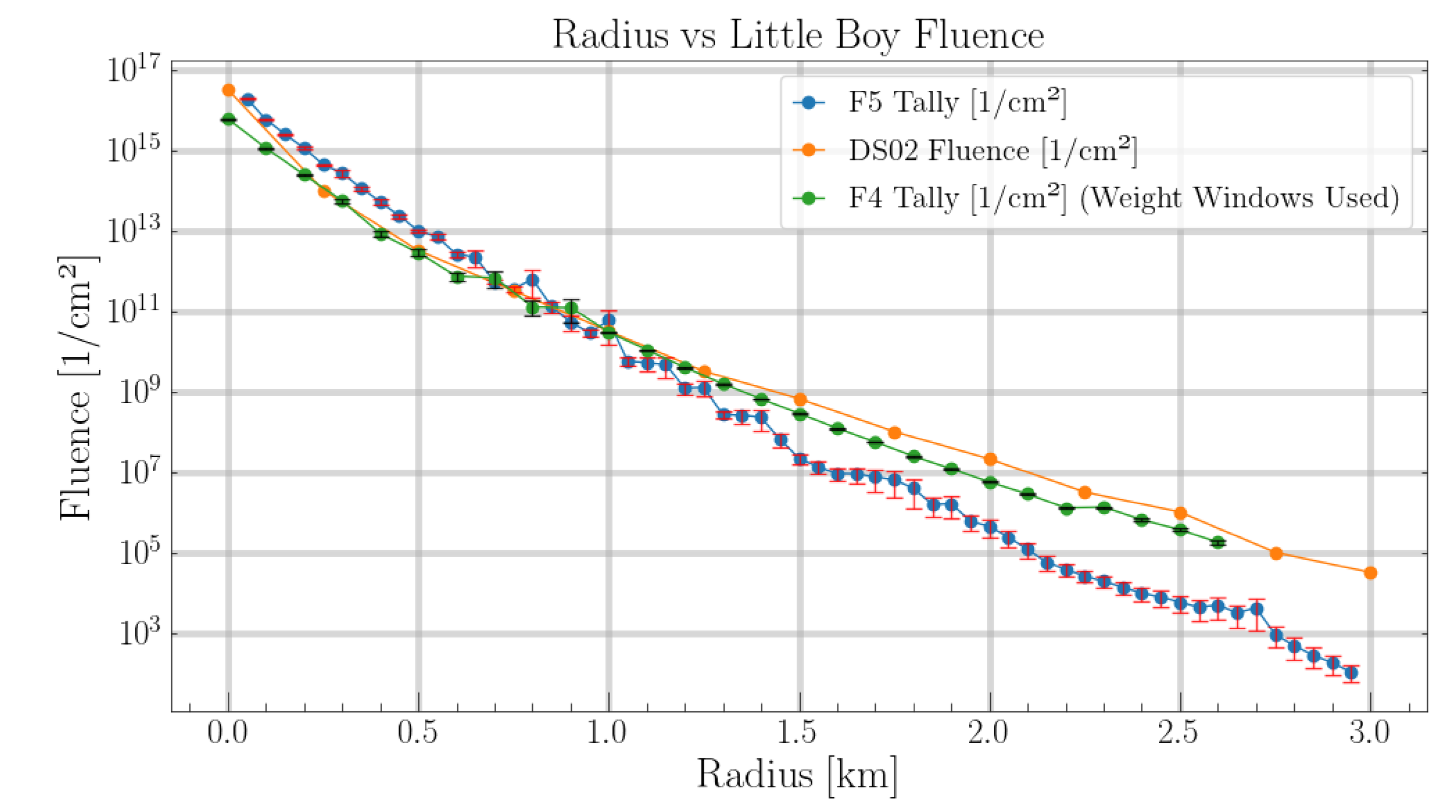



*Figure 4. Fluence vs radius for Little Boy at height of burst (600m) calculated by various variance reduction methods.*

Figure 4 compares the different ways to calculate fluence against the Dosimetry System 2002 fluence. The F4 tally and F5 tally both calculate fluence but are fundamentally different tallies therefore will produce different results.

| F4 Tally | F5 Tally |
|---|---|
| Monte Carlo transport | Semi-deterministic transport |
| Large uncertainty | Fast convergence of standard deviation |
| Better representation of physics occurring at the desired point | Does not count secondary particle production or scattering |

## CONCLUSIONS

This work models results from DS02 using most recent methods and libraries which aids validation and verification efforts. In order to meet that objective, this work successfully:

- Converted DS02 source into MCNP source terms and verified consistency
- Implemented mesh based weight windows to reduce statical error at large distances
- Illustrated benefits of F4 tallies with weight window

### Future work

- Calculate dose to compare transport models against other models
  - Dosimetry System 2002, Mercury
- Investigate the effectiveness of other variance reduction techniques
  - DXTRAN, energy weight windows
- Repeat for Little Boy photons and Fat Man neutrons/photons

## CONTACT

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[1] *MCNP 6.3.0 Theory & User Manual.* Kulesza, Joel A, et al. 2022

[2] *Standardized Unclassified Little Boy and Fat Man Outputs.* Richard L. Holmes and Stephen W. White. 2013

[3] *Dosimetry System 2002.* Ch 2 & 3. Stephen W. White, Robert T. Santoro, et al. 2002

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